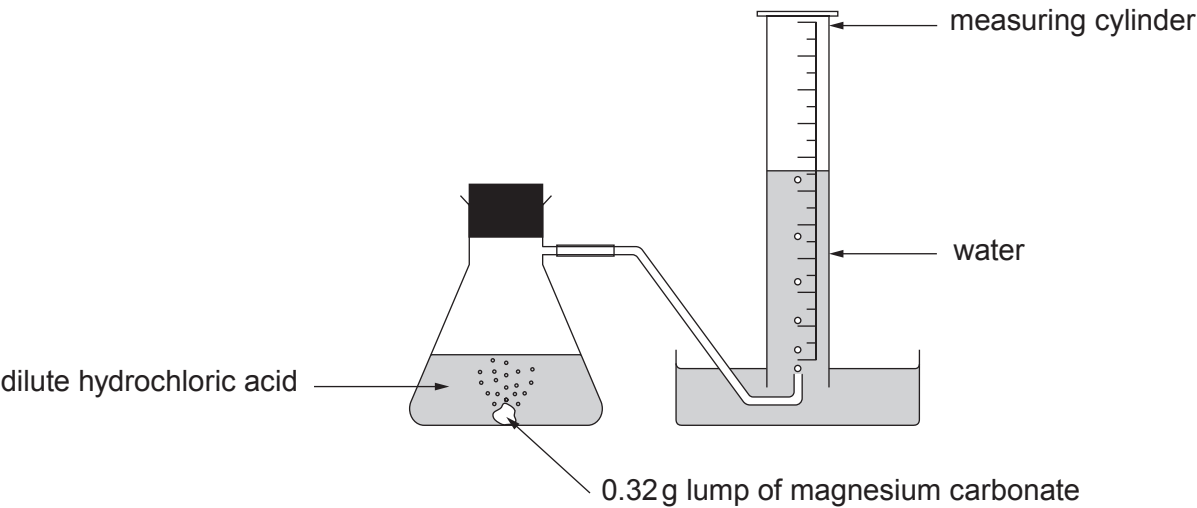


2. A student carried out an experiment to investigate the speed of the reaction between a **lump** of magnesium carbonate of mass 0.32 g and excess dilute hydrochloric acid at 20 °C.

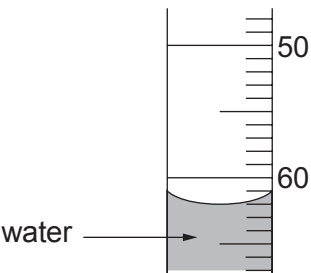
Magnesium carbonate reacts with dilute hydrochloric acid forming carbon dioxide gas. The total volume of carbon dioxide formed was recorded every 5 minutes for 40 minutes.



The results are shown below. The result for 15 minutes is missing.

Time (minutes)	0	5	10	15	20	25	30	35	40
Volume of carbon dioxide formed (cm ³)	0	20	41		79	83	90	90	90

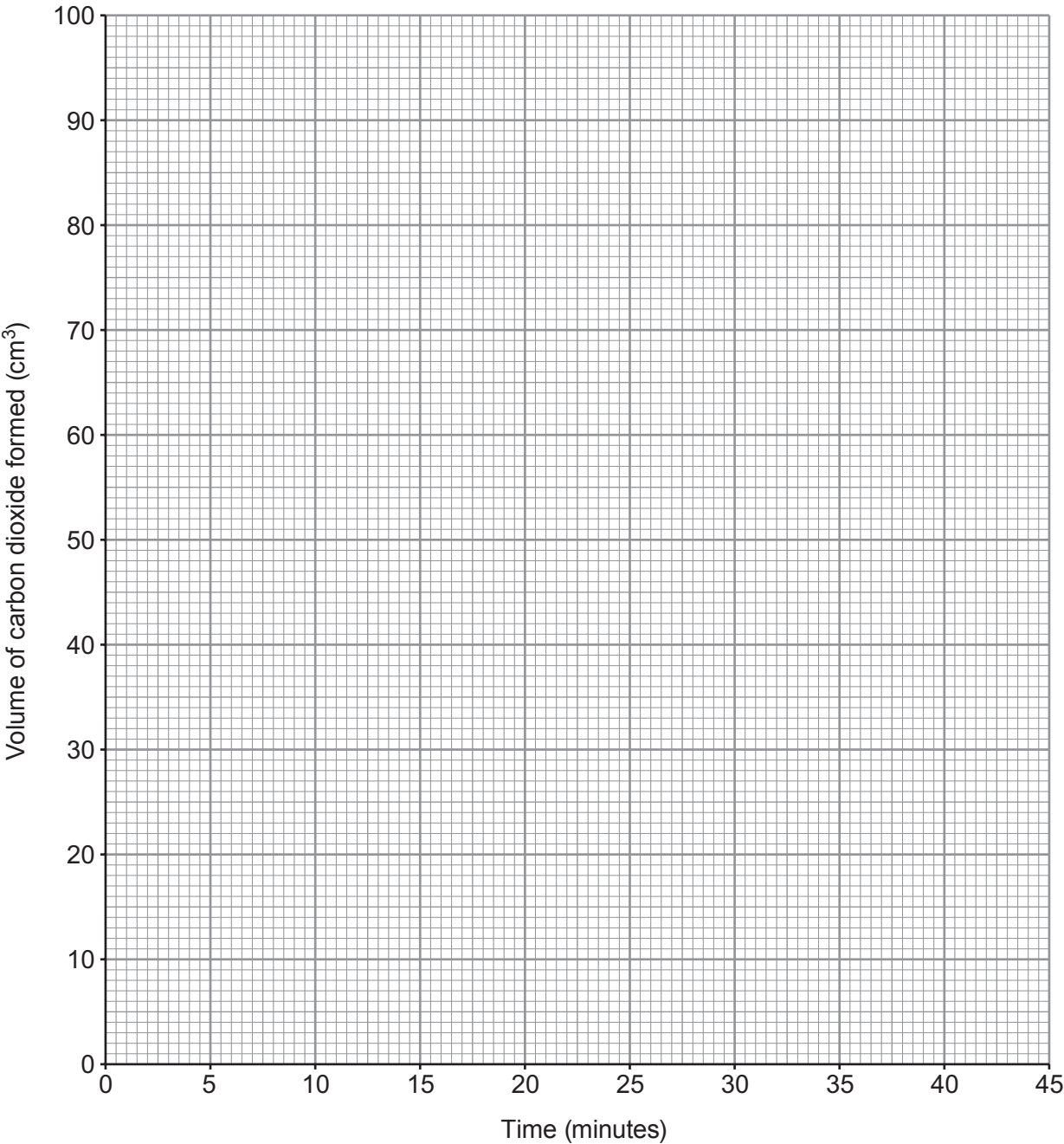
- (a) Use the diagram below to find the volume of carbon dioxide gas formed after 15 minutes. [1]



volume of carbon dioxide = cm³

- (b) Plot the results from the table, including your answer to part (a), on the grid below. Draw a suitable line and **label this graph X**.

[3]



- (c) Sketch the graph you would expect if the experiment were repeated using 0.32 g of magnesium carbonate **powder** instead of the lump of magnesium carbonate. **Label this graph Y**.

[2]

(d) State and explain, using particle theory, the effect of **increasing** the concentration of the hydrochloric acid. [3]

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.....

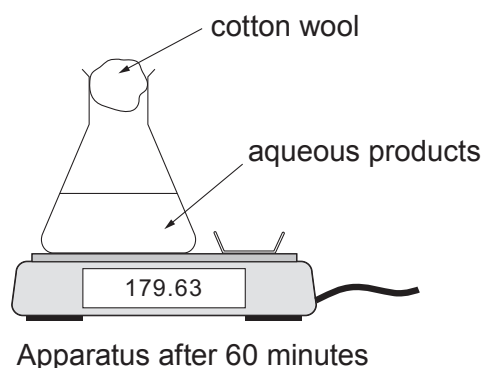
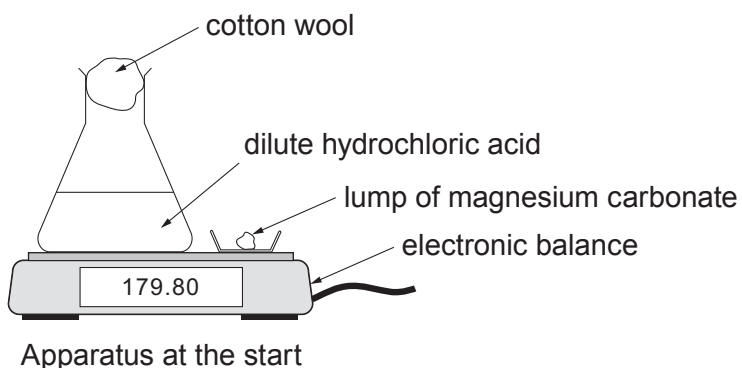
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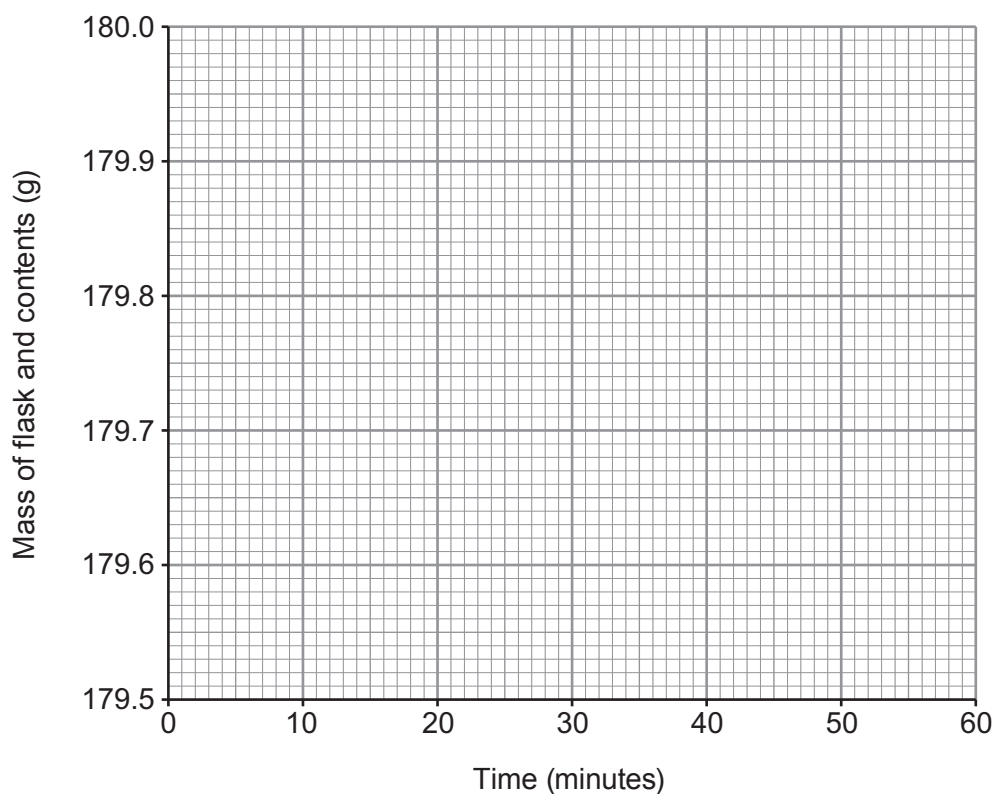
- (e) The student investigated the same reaction using a different apparatus. A lump of magnesium carbonate was added to excess dilute hydrochloric acid at 20 °C.



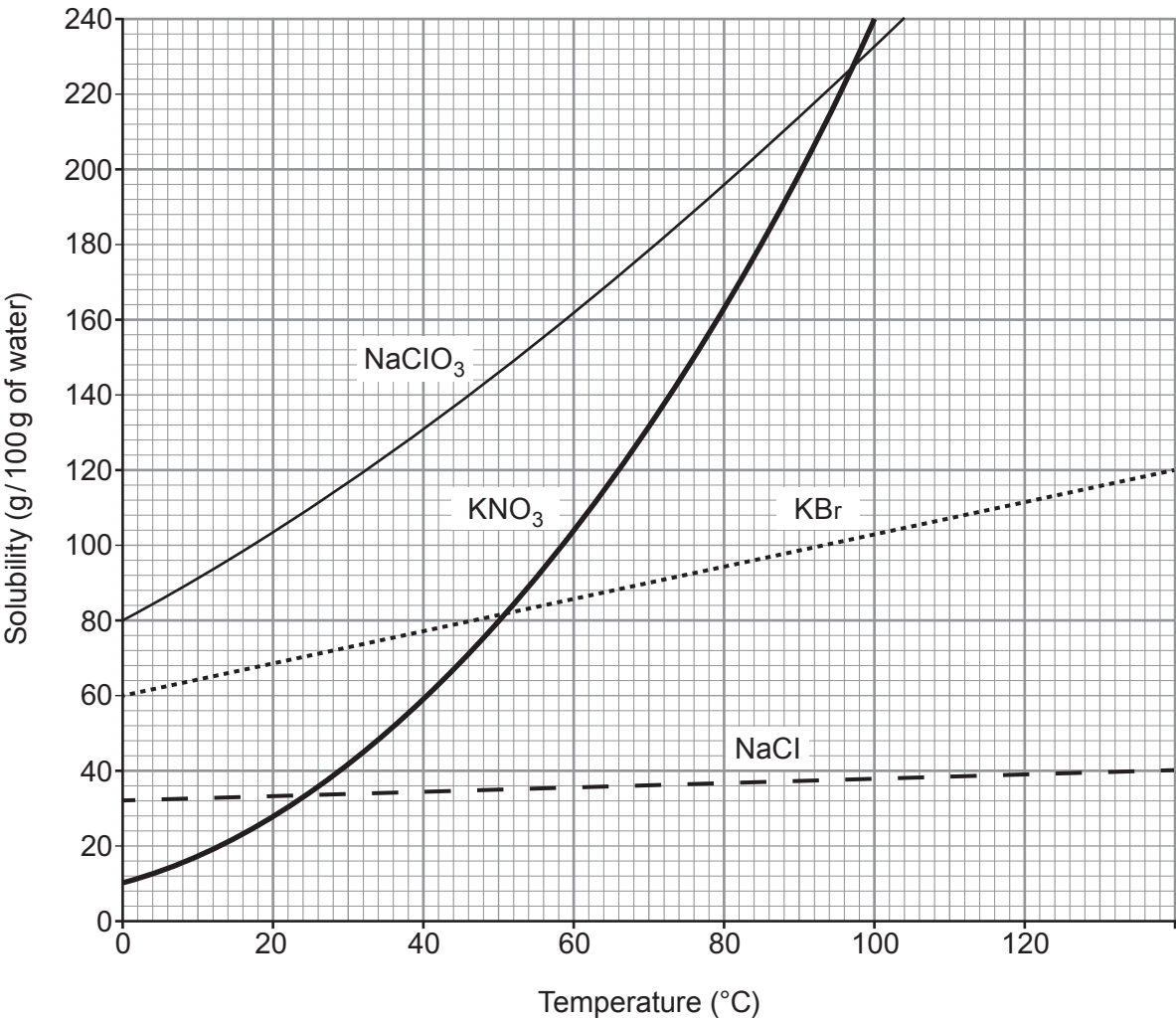
The change in mass was recorded for 60 minutes and displayed as a graph on a computer screen. The reaction took 40 minutes to complete.

Sketch the graph you would expect to see.

[2]



2. The grid below shows the solubility curves for four ionic compounds.



- NaClO₃ sodium chlorate
- KNO₃ potassium nitrate
- KBr potassium bromide
- NaCl sodium chloride



- (a) (i) Give the temperature at which the solubility of potassium nitrate and potassium bromide is the same. [1]

..... °C

- (ii) Calculate the mass of solid potassium nitrate that would form if a saturated solution in 200 g of water were cooled from 100 °C to 20 °C. [3]

Mass = g

- (iii) Suggest why a student may be surprised at the temperature range shown on the solubility curves. [1]

.....
.....

- (b) (i) Give the symbols of the **ions** of Group 1 elements present in the compounds shown on the grid. [1]

.....

- (ii) Explain how these ions are formed from their atoms. [2]

.....
.....

- (c) Potassium nitrate reacts with aluminium hydroxide to produce aluminium nitrate and potassium hydroxide.

Balance the symbol equation for the reaction taking place. [1]



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only

(c) Tick (✓) the box which gives **one** definite conclusion that can be drawn using **only** the data in **Figure 2**. [1]

Fluoridation has no effect on levels of decay

☐

People have reduced their intake of sugary foods over this period

☐

More than one factor affects levels of decay

☐

Fluoridation is the main cause of falling levels of decay

☐

(d) ‘Mass medication’ is an argument often given to oppose fluoridation of water supplies. Explain what is meant by the term *mass medication*. [1]

.....

.....

.....

.....

END OF PAPER

6



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		





THE PERIODIC TABLE

Group 1 2 3 4 5 6 7 0

1 H Hydrogen 1																				4 He Helium 2

1

H

Hydrogen

1

Key

A_r

Symbol

Name

Z

relative atomic mass

atomic number

Surname	Centre Number	Candidate Number
Other Names		0

**GCSE**

3410UA0-1



S19-3410UA0-1

WEDNESDAY, 12 JUNE 2019 – MORNING

CHEMISTRY – Unit 1:
Chemical Substances, Reactions and
Essential Resources

HIGHER TIER

1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this examination paper
 you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen
 or correction fluid.

Write your name, centre number and candidate number
 in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet.
 If you run out of space, use the additional page at the back
 of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **9** is a quality of extended response (QER) question where your writing skills will be assessed.

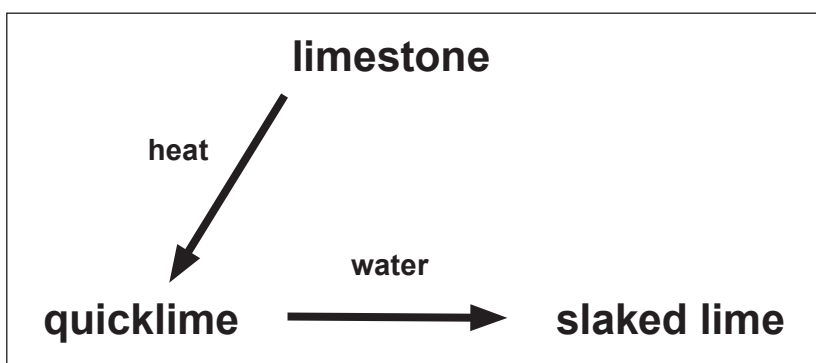
The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	7	
2.	7	
3.	9	
4.	6	
5.	7	
6.	7	
7.	9	
8.	12	
9.	6	
10.	10	
Total	80	

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2. The following diagram shows how limestone, CaCO_3 , can be converted into useful products.



(a) When a piece of limestone is heated strongly its mass decreases.

State the type of reaction taking place.

[1]

.....

(b) (i) Describe what is **seen** when limestone is heated and converted into quicklime. [1]

.....

(ii) Write a balanced symbol equation for the reaction taking place.

[2]

.....

(c) (i) Describe what is observed when quicklime is converted into slaked lime. [1]

.....

(ii) Write a balanced symbol equation for the reaction taking place.

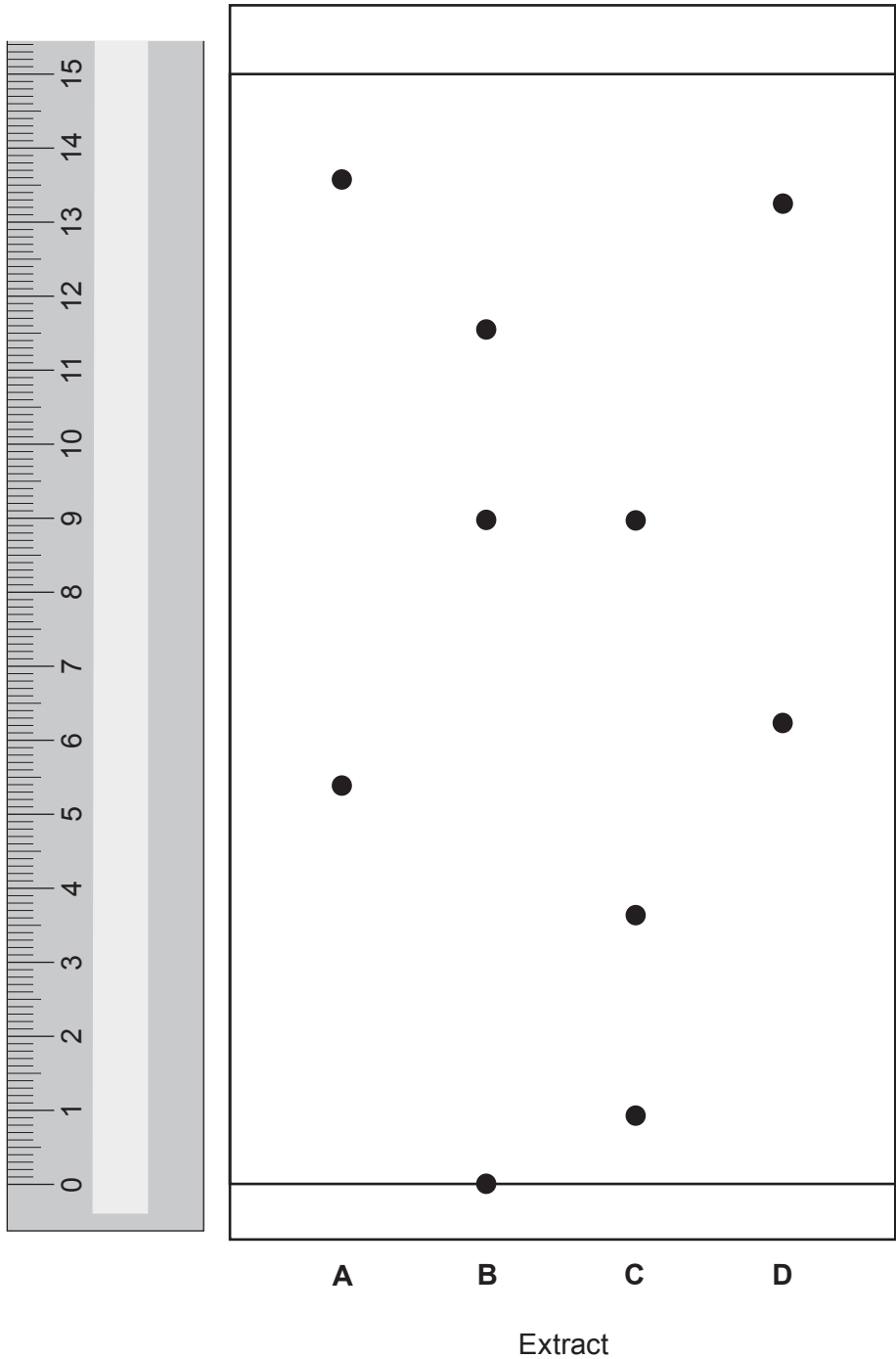
[2]

.....



2. Paper chromatography can be used to identify plant leaf pigments. This process uses a chemical called acetone as the solvent instead of water.

The diagram shows the chromatogram of plant leaf extracts **A**, **B**, **C** and **D** in acetone.



Examiner
only

(a) All of the extracts contain a mixture of pigments with different R_f values.

For which plant leaf extract, **A**, **B**, **C** or **D**, is the **highest** R_f value 0.60?

Give your reasoning. [3]

Extract

Reasoning

.....
.....

(b) Explain why the pigments travel different distances on the chromatogram. [2]

.....
.....
.....

(c) One of the extracts contains a pigment which is insoluble in acetone.

State the **letter** of this extract. Explain your choice. [2]

Extract

Explanation

.....

(d) The chemical formula of the solvent acetone is C_3H_6O . Calculate the percentage by mass of carbon in acetone.

The relative formula mass (M_r) of acetone is 58. [2]

$A_r(C) = 12$

Percentage = %

9

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05



2. (a) The table shows information about some Group 1 elements.

Element	Relative atomic mass	Number of electrons in the outer shell	Melting point (°C)	Boiling point (°C)	Density (g/cm ³)
lithium	7	1	180	1342	0.53
sodium	23	1	98	883	0.97
potassium	39	1	63	759	0.89
rubidium	85	1	39	688	1.53
caesium	134	1	29	671	1.93

Use the information in the table to answer parts (i) and (ii).

(i) State the information which explains why the elements have similar chemical properties. [1]

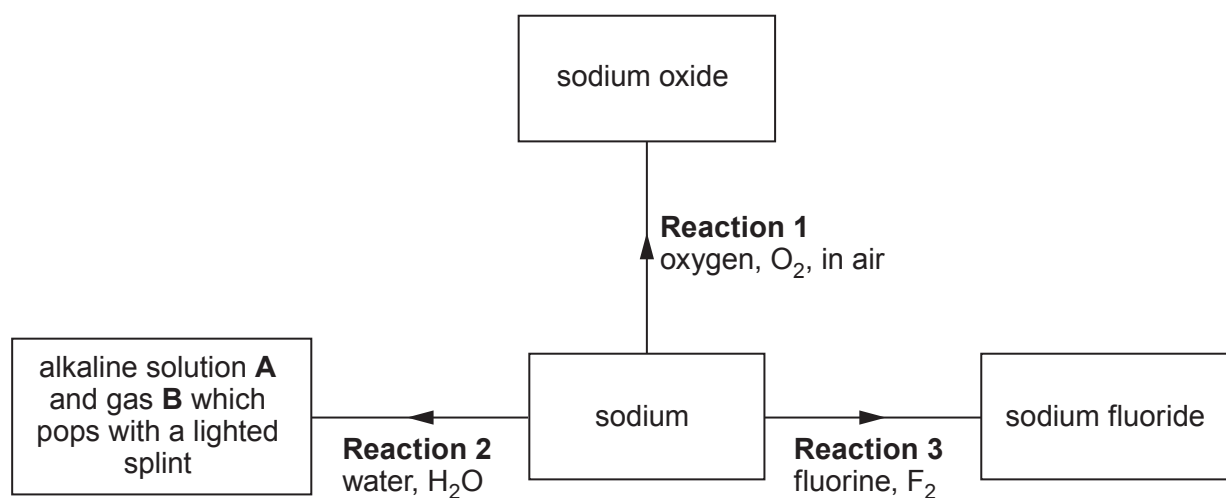
.....

(ii) State which **property** has a value which does **not** fit the trend down the group. [1]

.....



(b) The flow diagram shows some reactions of sodium.



(i) State how **Reaction 1** is prevented when storing sodium in the laboratory. [1]

.....

(ii) Give the names of alkaline solution **A** and gas **B**. [2]

..... and

(iii) Name the Group 1 metal which would react **least** violently with water. [1]

.....

(iv) Complete the symbol equation for **Reaction 3**. [1]

